

2-Mark Questions & Answers — DBMS Module 2: Relational Algebra & Calculus

Q1. What is Relational Algebra?

Relational Algebra is a procedural query language that takes one or two relations as input and produces a new relation as output using mathematical operators like Selection (σ), Projection (π), Union ($\cup$), and Join ($\bowtie$).

Q2. Differentiate between Selection (σ) and Projection (π) operations.

- **Selection (σ):** Horizontal filtering operation that selects rows (tuples) that satisfy a specified predicate condition.
- **Projection (π):** Vertical filtering operation that selects specific columns (attributes) and removes duplicate tuples.

Q3. What is Union Compatibility in Relational Algebra?

Two relations R and S are **Union Compatible** if:

1. Both relations have the **same number of attributes** (same arity).
2. The domains of corresponding attributes are identical in data type and order.

Q4. Define Cartesian Product ($\times$) with an example.

The **Cartesian Product ($R \times S$)** combines every tuple of relation R with every tuple of relation S. If R has n tuples and S has m tuples, $R \times S$ contains $n \times m$ tuples.

Q5. What is Tuple Relational Calculus (TRC)?

Tuple Relational Calculus (TRC) is a non-procedural query language where queries are expressed in the mathematical logic form:

$$\{ t \mid P(t) \}$$

where t is a resulting tuple variable and $P(t)$ is a predicate formula specifying conditions the tuple must satisfy.